Koq

SESSION 2024/2025 SEMESTER 1

ULRF/UKQF TEKNOLOGI DRON (DRONE TECHNOLOGY)

Reflection Report on the Drone Club's Service-Learning Event Using the STARR Framework

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Reflection Report Using the STARR Framework

1. Situation

As a member of the DRONE TECHNOLOGY, I participated in several activities that enhanced my understanding of drone technology and my ability to guide others. These activities included:

1. Class Challenge: I explored the Tello Edu drone using both remote control for manual navigation and block programming for automated flight tasks. The goal was to navigate the drone around obstacles and explore its capabilities in both manual and programmed modes.
2. Service-Learning Event: During the "The Future is Flying: A Drone Exploration" event, I served as a person in charge in activity unit, teaching participants how to operate drones using remote control and assisting them in understanding basic drone functionalities.
3. Drone Knowledge Session**:** In a classroom setting, we learned from an instructor about the theoretical and practical aspects of drones, including flight mechanics, safety protocols, and basic operations.
4. DJI Mini 3 Aerial Exploration**:** I practiced flying the DJI Mini 3 drone to capture high-altitude photographs of UTM's landmarks, experiencing broader perspectives and learning precision control.
5. CESCO 4.0 Exhibition**:** At UTM DSI, we showcased our service-learning project during the CESCO 4.0 event by setting up a booth and engaging visitors with demonstrations of our drone activities.

2. Task

1. **Class Challenge:**
    - Use **remote control** to manually navigate the Tello Edu drone through an obstacle course.
    - Transition to **block programming** to automate the drone's flight, navigating around pillars and reaching a target destination.
2. **Service-Learning Event:**
    - Guide participants in operating drones using remote control, focusing on takeoff, hovering, and landing.
    - Troubleshoot issues participants faced during drone operation and provide real-time support.
3. **Drone Knowledge Session:**
    - Learn the fundamental principles of drone operation, including flight mechanics and safety procedures.
    - Apply theoretical knowledge during practice sessions.
4. **DJI Mini 3 Aerial Exploration:**
    - Fly the DJI Mini 3 drone to explore UTM’s wider landscape and capture high-altitude photos of its landmarks.

    - Ensure stable flight and precise control while adhering to safety protocols.
5. **CESCO 4.0 Exhibition:**
    - Assist in setting up a booth to showcase our service-learning activities.
    - Demonstrate Tello Edu programming and DJI Mini 3 aerial photography, engaging visitors and answering questions about drone applications.

3. Action

1. **Class Challenge:**
    - I manually piloted the Tello Edu drone using its remote control, focusing on precision and responsiveness while navigating obstacles.
    - I then programmed the drone using block programming, testing and refining commands to ensure the drone completed the obstacle course autonomously.
2. **Service-Learning Event:**
    - I guided participants in using the Tello Edu drone for basic operations, explaining remote control features step-by-step.
    - I addressed participants' challenges, such as drone instability or difficulty in takeoff, ensuring a smooth and engaging experience.
3. **Drone Knowledge Session:**
    - I actively participated in the session and practicing basic drone maneuvers during the hands-on component.
4. **DJI Mini 3 Aerial Exploration:**
    - I flew the DJI Mini 3 drone to various landmarks within UTM, capturing high-quality aerial photos and practicing stable flight at higher altitudes.
    - I adjusted the drone's angles and movements to get the best perspectives while ensuring safety compliance.
5. **CESCO 4.0 Exhibition:**
    - I helped set up the booth with posters, drone displays, and photos from our activities.
    - I engaged visitors by demonstrating drone operations and explaining the objectives and outcomes of our service-learning activities.

4. Results

1. **Class Challenge:**
    - Successfully navigated the Tello Edu drone manually and programmed it to autonomously complete the obstacle course.
    - Improved my understanding of drone navigation and programming.
2. **Service-Learning Event:**
    - Participants gained hands-on experience and successfully operated drones with my guidance.
    - Feedback highlighted the interactive and engaging nature of the event.
3. **Drone Knowledge Session:**
    - Gained a solid theoretical foundation and practical understanding of drone operation and safety protocols.

4. **DJI Mini 3 Aerial Exploration:**
    - Captured stunning aerial photos of UTM's landmarks, showcasing the campus from a broader perspective.
    - Improved my skills in high-altitude drone control and stability management.
5. **CESCO 4.0 Exhibition:**
    - The booth attracted significant attention, with visitors appreciating the demonstrations and learning about the impact of our service-learning project.
    - My presentations effectively communicated the practical applications of drone technology.

5. Reflection (Relating and Refining)
These activities provided a comprehensive learning experience, enhancing both my technical skills and my ability to communicate effectively about drones.

Class Challenge: Combining manual control and block programming taught me the importance of iterative problem-solving and adapting to different flight scenarios.

Service-Learning Event: I learned the value of patience and clear communication in guiding others.

Drone Knowledge Session: Theoretical knowledge reinforced my practical skills, improving my confidence in drone operations.

DJI Mini 3 Exploration: High-altitude flights improved my precision and awareness of drone safety.

CESCO 4.0 Exhibition: Presenting our activities to a broad audience honed my engagement and presentation skills.

Key takeaways:

Combining Methods: Learning both manual and programmed control of drones enhances versatility in operations.

Engagement Skills: Clear instructions and hands-on demonstrations foster greater interest and understanding.

Safety First: Diligence in following safety protocols ensures successful operations, especially in high-altitude scenarios.

Moving forward, I plan to:

Explore more advanced programming techniques to tackle complex drone challenges.

Practice high-altitude drone operations to improve stability and precision further.

Refine my ability to simplify technical concepts for a broader audience.

These experiences significantly strengthened my technical expertise, teamwork, and communication skills, preparing me for future drone-related initiatives.